Name	Subject Class	Class	Candidate Number
	2BI		

ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY **9744/02**
17 AUGUST 2018
HIGHER 2 **2 hours**

Paper 2 Structured Questions

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Total	100

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on the top of this page.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
No additional materials are required.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's use

1 Cholesterol is synthesised in the smooth endoplasmic reticulum (SER) in liver cells by a series of enzyme-catalysed reactions. Cholesterol is then transported to the Golgi apparatus where they are packaged into vesicles and subsequently released into a membrane-bound duct of the liver.

Fig. 1.1 is an electron micrograph of a section of a liver tissue.

Fig. 1.1

(a) Name structure **T** in Fig. 1.1 and describe its role in liver cells.

[2]

(b) Both prokaryotes and structure **T** have membrane proteins to help them perform the role described in (a). Suggest how prokaryotes perform this role.

[3]

For examiner's use

(c) Describe the role of cholesterol in the cell surface membrane.

[1]

(d) Suggest how cholesterol is transported from the Golgi apparatus to the membrane-bound duct of the liver.

[2]

[Total: 8]

2 Fig. 2.1 shows the activation of a G-protein linked receptor (GPLR) found in the cell surface membrane upon binding of a drug, salbutamol. Salbutamol is often used in the treatment of asthma – a long-term condition which narrows the airways in the lungs. The G protein consists of three subunits - α, β, and γ.

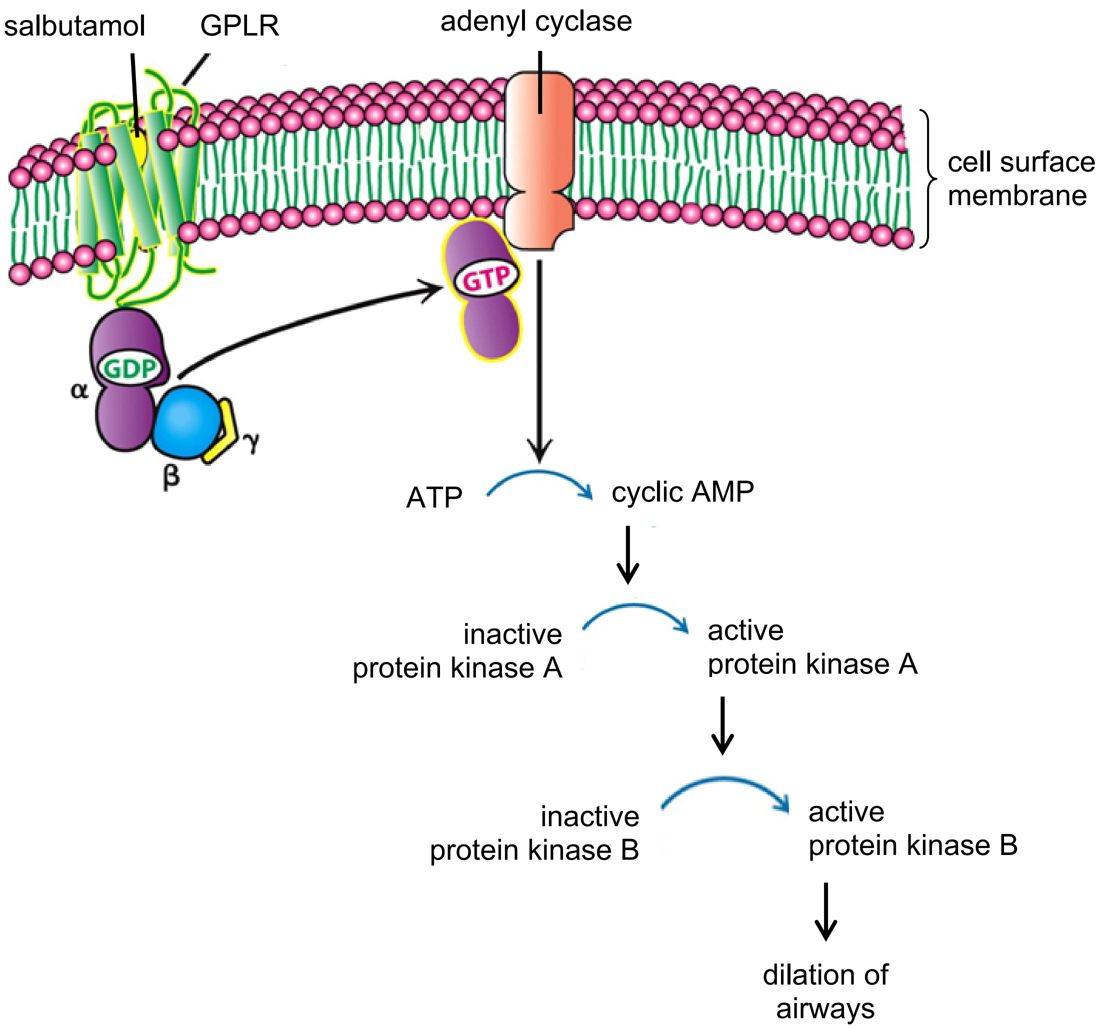

Fig. 2.1

(a) Describe how GPLR is held within the cell surface membrane.

....

....

....

.... [2]

For examiner's use

(b) With reference to Fig. 2.1, describe how the use of salbutamol can relieve symptoms of asthma.

[4]

(c) There are many different types of receptors used by a cell to transduce extracellular signals. Some are located on the cell surface membrane while others are located within the cell.

Explain why some receptors are found on the cell surface membrane while others are found within the cell.

[2]

[Total: 8]

For examiner's use

3 tRNA molecules have an important role in gene expression. They can be found in the cytoplasm, mitochondria and chloroplasts of eukaryotic cells. Fig. 3.1 shows the structure of a tRNA molecule that is able to carry the amino acid threonine in the cytoplasm of a eukaryotic cell. It is 77 ribonucleotides long.

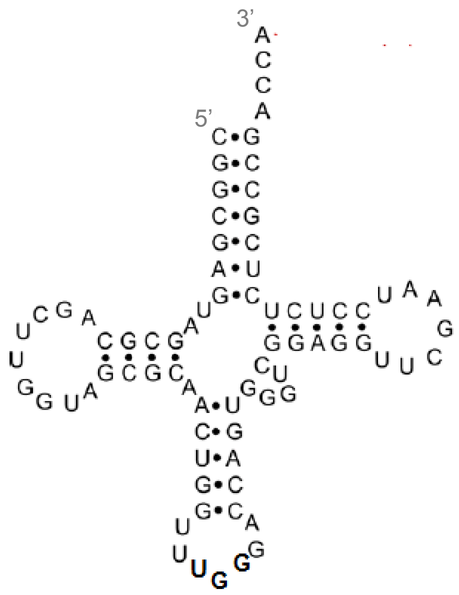

Fig. 3.1

(a) **(i)** With reference to Fig. 3.1, describe how the structure of a tRNA molecule differs from the structure of a mRNA molecule.

....

....

....

.... [2]

(ii) Explain how these differences allow tRNA and mRNA molecules to perform their roles.

....

....

....

.... [2]

Fig. 3.2 shows a tRNA molecule that is able to carry the amino acid threonine in mitochondria. It is 62 ribonucleotides long.

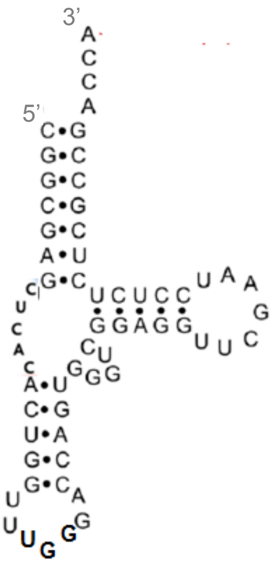

Fig. 3.2

(b) The structures of the tRNA molecules in the mitochondria and the cytoplasm of a eukaryotic cell are different although they are both able to carry the amino acid threonine. Suggest a reason for this difference.

....

....

....

.... [2]

There are 61 unique tRNA molecules in the cytoplasm of a eukaryotic cell.

(c) **(i)** Explain why there are 61 unique tRNA molecules in the eukaryotic cell.

....

....

....

.... [2]

For examiner's use

(ii) Scientists found that there are more than 120 gene loci coding for tRNA molecules in the nuclear DNA, many of which are duplicate copies of the same tRNA gene.

Suggest an advantage of having more than 120 gene loci coding for the 61 tRNA molecules in the eukaryotic cell.

....

.... [1]

(d) Explain how the structure of RNA polymerase allows for the synthesis of tRNA molecules.

....

....

....

....

....

.... [3]

[Total: 12]

For examiner's use

4 Fig. 4.1 shows the reproductive cycles of a temperate bacteriophage.

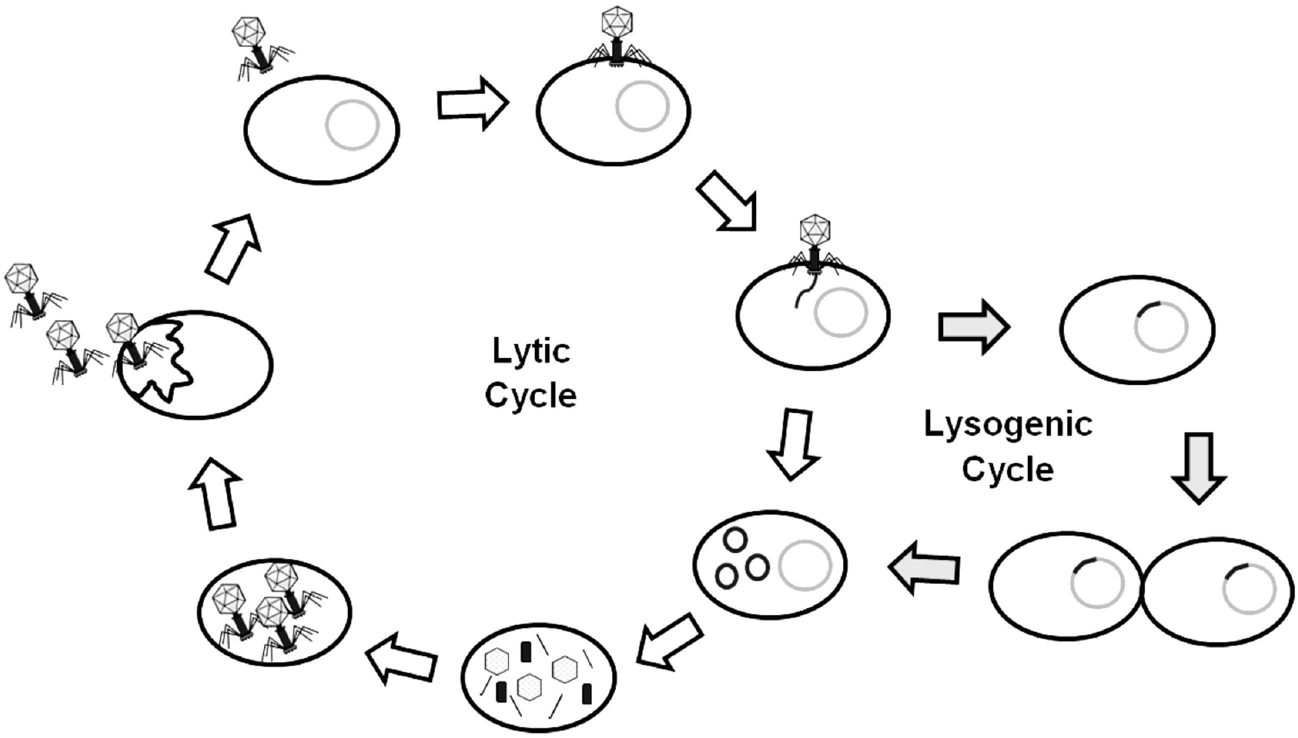

Fig. 4.1

(a) With reference to Fig. 4.1, describe the roles of the bacteriophage structures in its reproductive cycles.

....

....

....

....

....

....

....

.... [4]

For examiner's use

(b) While their prokaryotic hosts contain both DNA and RNA, bacteriophages contain only DNA. Suggest why it is sufficient for bacteriophages to contain only DNA for their reproductive cycles.

[2]

(c) Describe an advantage of the lysogenic cycle of a bacteriophage to a population of bacteria.

[2]

[Total: 8]

For examiner's use

5 Telomerase reverse transcriptase (TERT) is a component of the enzyme telomerase, which is responsible for maintaining telomere lengths in epidermal stem cells. These cells will differentiate to form specialised skin cells such as keratinocytes**.** A study on the role of the regulatory protein c-MYC in the synthesis of TERT mRNA was conducted in these cells. Fig. 5.1 shows the relative levels of TERT mRNA and c-MYC mRNA.

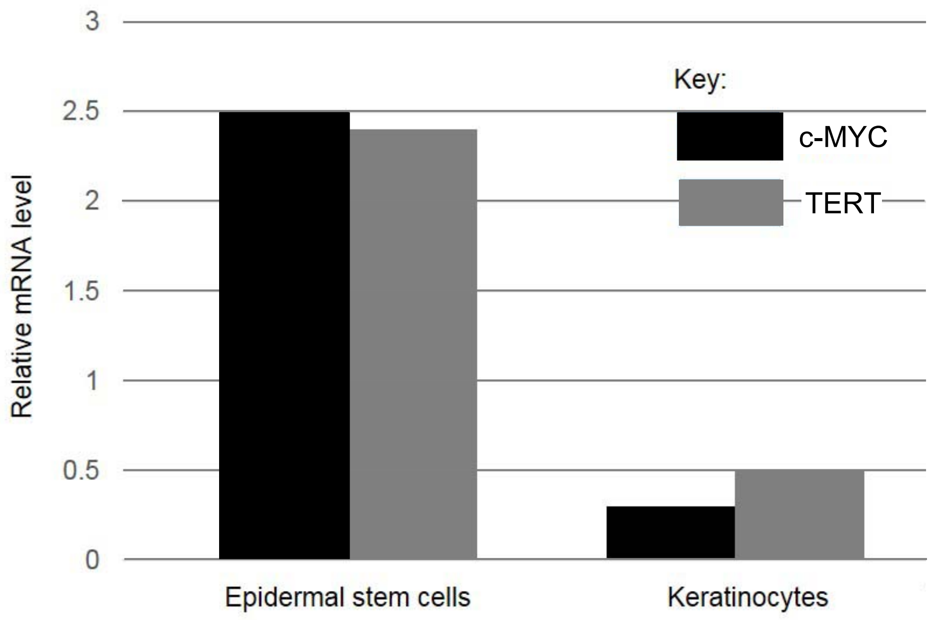

Fig. 5.1

(a) (i) With reference to Fig. 5.1, describe the relationship between the relative levels of TERT mRNA and c-MYC mRNA.

....

....

....

.... [2]

(ii) Suggest a role of the c-MYC protein in the regulation of *TERT* gene expression.

....

....

....

....

....

.... [3]

(b) Explain why TERT has a significant role in epidermal stem cells.

[3]

(c) Suggest why *TERT* gene is absent in bacteria.

[2]

[Total: 10]

For examiner's use

6 Many diseases arise due to gene mutations and/or chromosomal aberrations.

(a) Distinguish between gene mutations and chromosomal aberrations.

....

....

....

.... [2]

Fig. 6.1 shows an error occurring during meiosis II.

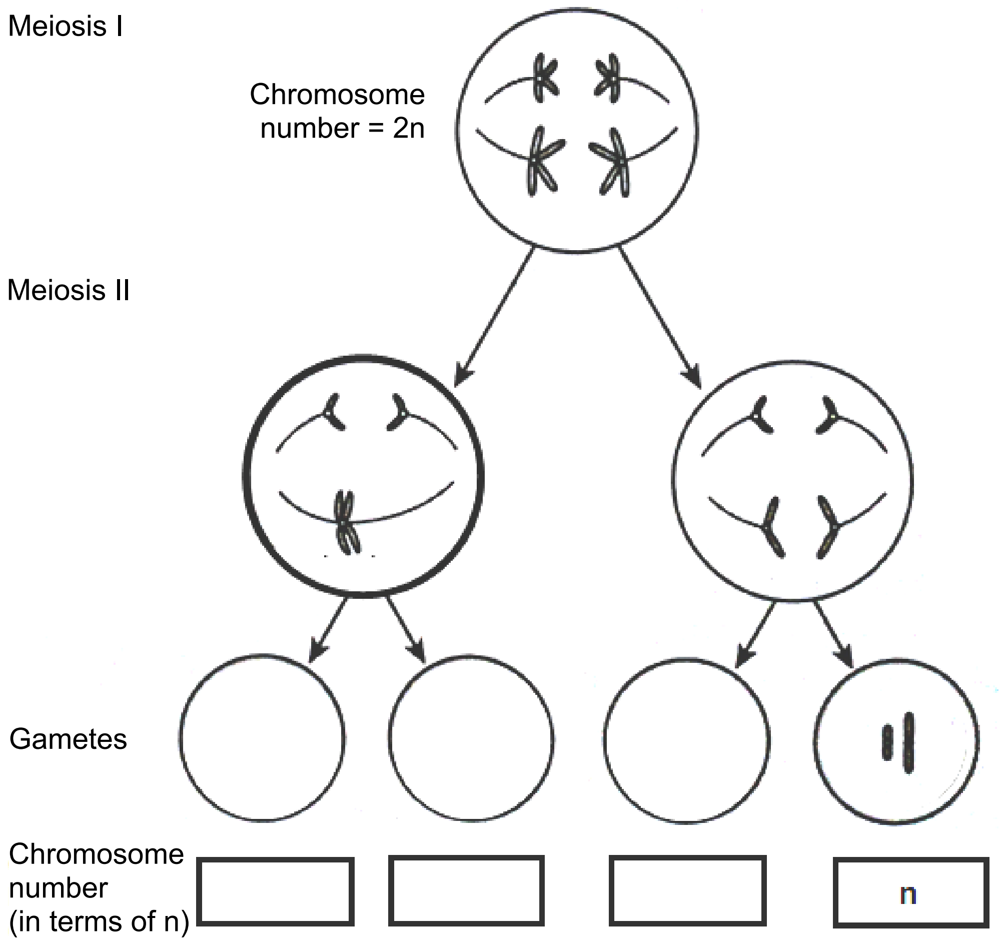

Fig. 6.1

(b) **(i)** Complete Fig. 6.1 with the correct chromosome structures and chromosome number (in terms of n) in the gametes. [2]

(ii) Explain the error shown in Fig. 6.1.

[2]

(iii) Predict the chromosomal number (in terms of n) in each of the four gametes if a similar error had occurred in meiosis I instead of meiosis II.

[1]

(c) Such errors can occur during mitosis too. Comment on whether this would lead to changes in the gene pool of a population.

[1]

[Total: 8]

For examiner's use

7 **(a)** Explain how genotype is linked to phenotype.

[3]

In the fruit fly, *Drosophila*, a gene determines the appearance of a dark transverse band on the thorax. Another gene determines the presence of crossveins on the wings. Two pure-breeding *Drosophila* were crossed, where one is banded with crossveins on the wings while the other is non-banded without crossveins on the wings. The F1 offspring were then test-crossed to produce offspring in the following ratio:

banded without crossveins	21
banded with crossveins	483
non-banded without crossveins	512
non-banded with crossveins	35

(b) Using the symbols **B** for non-banded thorax and **b** for banded thorax, **D** for presence of crossveins and **d** for absence of crossveins, draw a genetic diagram to show the results of the test-cross.

[4]

(c) Explain how the new combination of alleles arose in the gametes of the test-cross in part (b).

[3]

(d) Distinguish sex linkage from the mode of inheritance of the two genes observed in *Drosophila*.

[2]

[Total: 12]

For examiner's use

8 Concentrations of ribulose bisphosphate (RuBP) and reduced NADP were measured in samples of actively photosynthesising green algae in an experimental chamber. CO_2 is supplied continuously into the chamber throughout the experiment.

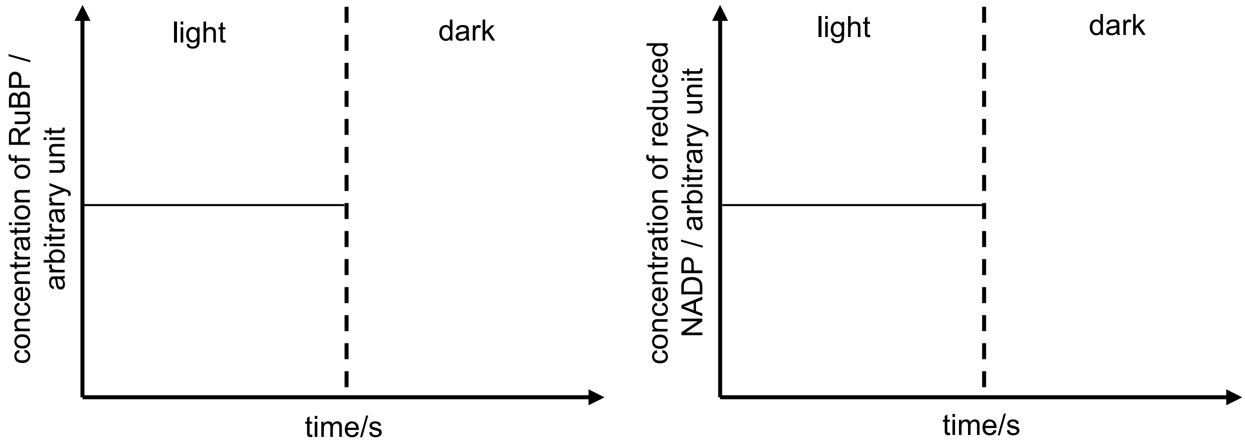

Fig. 8.1

(a) (i) In Fig. 8.1, complete each graph to show the concentrations of RuBP and reduced NADP when the light source was turned off. [2]

(ii) Explain the shape of the two graphs when the light source was turned off.

....

....

....

....

....

....

....

.... [4]

For examiner's use

(b) 'Calvin cycle is the reverse of Krebs cycle'. Comment on this statement.

[3]

(c) The Krebs cycle must be carefully regulated by the cell. If it is permitted to proceed unchecked, large amounts of metabolic energy would be wasted in the over-production of reduced coenzymes and ATP. α-ketoglutarate dehydrogenase is an enzyme in the Krebs cycle which catalyses the conversion of α-ketoglutarate to succinyl coA.

Suggest how reduced NAD can inhibit the action of α-ketoglutarate dehydrogenase.

[3]

[Total: 12]

9 A study reported on the change in beetle sizes in a population due to climate change. Larger beetles are affected when compared to smaller beetles. This could be because larger beetles are less likely to get enough oxygen to sustain a higher metabolic rate caused by higher atmospheric temperatures. Fig. 9.1 shows the effect of a temperature rise of 2°C on the size of larger and smaller beetles in a population over 100 generations.

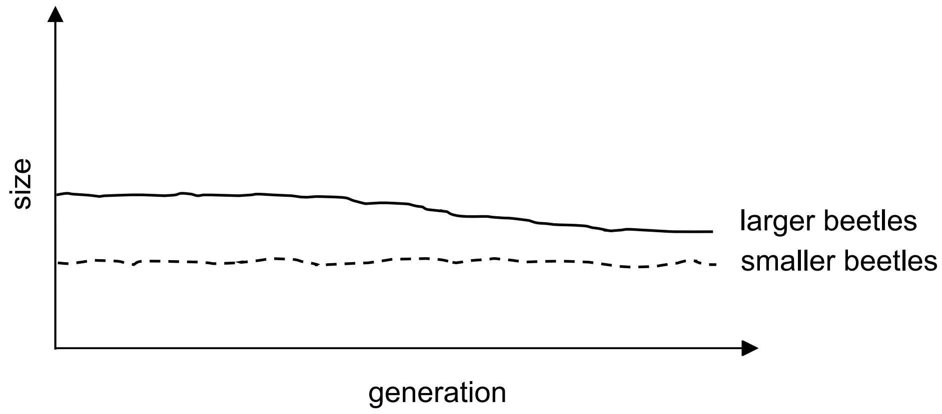

Fig. 9.1

(a) Explain how a temperature rise of 2°C may cause a change in the gene pool of the population of beetles over time.

[4]

(b) Suggest why a speciation event is unlikely to occur in this population of beetles as a result of climate change.

[2]

For examiner's use

(c) Suggest the consequences of a decrease in beetle size on the ecosystem.

[2]

(d) Describe how molecular methods can be used to confirm that two beetles with similar morphology are of the same species.

[4]

[Total: 12]

For examiner's use

10 Multidrug-resistant tuberculosis (MDR-TB) is a form of tuberculosis (TB) caused by the bacterium *Mycobacterium tuberculosis,* which is resistant to treatment with the antibiotic rifampin. The first antibiotic treatment for TB began in 1943 and some strains *M. tuberculosis* developed resistance to rifampin through genetic changes. Fig. 10.1 shows rifampin binding to RNA polymerase in bacterial cells.

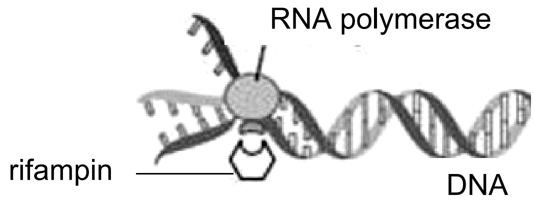

Fig. 10.1

(a) With reference to Fig. 10.1, explain why rifampin may be used to treat tuberculosis.

....

....

....

.... [2]

(b) Suggest how the genetic changes which caused rifampin resistance could be transferred within a population of *M. tuberculosis* through direct contact of bacteria cells.

....

....

....

....

....

.... [3]

Bacilli Calmette-Guérin (BCG) is a vaccine containing attenuated bacteria and can provide protection against tuberculosis. The attenuated bacteria have antigens which stimulate an immune response. Recent studies have shown that BCG can also give protection against leprosy, a disease caused by the bacterium *Mycobacterium leprae*.

(c) (i) Outline the roles of T lymphocytes in the adaptive immune response after the administration of BCG.

....

....

....

....

....

.... [3]

(ii) Suggest why BCG can also provide protection against leprosy.

....

....

....

.... [2]

[Total: 10]